

Sourav Chakraborty

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RESEARCH INTERESTS	Multi-Agent Reinforcement Learning (MARL), Bandit Algorithms, and Online Optimization under uncertainty, combining tools from statistics, optimization, and sequential decision theory. Developing scalable decision-making frameworks leveraging locality and structural constraints in large-scale networked systems.
EDUCATION	University of Colorado Boulder , Colorado, USA Ph.D. Computer Science, Dec 2026 (Expected). M.Sc. Computer Science, May 2022; GPA: 4.0. Thesis: <i>Incentivized Exploration in Non-stationary Stochastic Bandits</i> (thesis) Advisor: Lijun Chen Birla Institute of Technology, Mesra , Ranchi, India B.E. Information Technology, June 2016.
WORK EXPERIENCE	California Institute of Technology (Caltech) , Pasadena, CA, USA Visiting Student Researcher. 2026–Present - Department of Computing and Mathematical Sciences; Working with Profs. Steven Low and Adam Wierman; developing bandit/MARL theory toward scalable networked learning and optimization. University of Colorado Boulder , Colorado, USA Graduate Student Researcher & Instructor. 2019–Present - Conducted independent research in the fields of Multi-Agent Reinforcement Learning and online learning, bridging rigorous theoretical analysis with implementation. - Employed as Instructor of Record (2 semesters), Graduate TA (10+ semesters), and Departmental Lead TA (3 consecutive years), overseeing instructional operations and mentoring a cohort of 280+ staff members. Flipkart , Bangalore, India Software Engineer 2016–2019 - Engineered an end-to-end query suggestion pipeline serving 50M+ daily sessions, boosting query coverage by 3x and product discovery CTR by ~15% using Java and Cascading. - Developed 5+ ranking signals (recency, semantic similarity, etc.) improving top-5 retrieval precision by ~12% for a 10M-query benchmark. - Architected the BRaaS(Backup-as-a-Service) platform adopted by 20+ microservices, reducing manual recovery time by 40% and eliminating single-point-of-failure risks. - Collaborated across infrastructure and product-search teams on engineering efforts spanning discovery quality and system reliability at scale.
RESEARCH AND PUBLICATIONS	Papers Under Review (<i>*Denotes equal contribution</i>) S. Chakraborty*, A. K. Rege*, C. Monteleoni, L. Chen, PROVABLY EFFICIENT NETWORKED MARL WITH CORRELATED POLICIES, 2026. S. Chakraborty*, A. K. Rege*, C. Monteleoni, L. Chen, DECENTRALIZED LEARNING IN CONTINUOUS MULTI-AGENT BANDITS, 2026. S. Chakraborty*, A. K. Rege*, C. Monteleoni, L. Chen, LEARNING FROM LOCAL WALKS ON DYNAMIC GRAPHS WITH BANDIT FEEDBACK, 2026. J. Hu, W. Chen, S. Chakraborty, L. Chen, ON REGRET BOUNDS OF QUANTUM BANDIT ALGORITHMS, 2026. A. Ghriss, S. Chakraborty, KERNELIZED LINEAR ATTENTION: BREAKING THE CAPACITY WALL WITH SYMMETRIC CONES, 2026. (Efficient linear-attention mechanism for long-context sequence modeling in large language models.) Peer-Reviewed Conference Publications (<i>*Denotes equal contribution</i>)

	1. S. Chakraborty*, A. K. Rege*, C. Monteleoni, L. Chen, MULTI-AGENT LIPSCHITZ BANDITS, <i>International Conference on Artificial Intelligence and Statistics (AISTATS 2026)</i>, Tangier, Morocco. (arXiv: 2602.16965) 2. S. Chakraborty*, A. K. Rege*, C. Monteleoni, L. Chen, FLICKERING MULTI-ARMED BANDITS, <i>Learning for Dynamics & Control Conference (L4DC 2026)</i>, Los Angeles, USA. (Oral Presentation). (★ Best Paper Finalist, 1 of 3; Top 1%). (arXiv: 2602.17315) 3. S. Chakraborty*, A. K. Rege*, C. Monteleoni, L. Chen, A UNIFIED FRAMEWORK FOR LOCALITY IN SCALABLE MARL, <i>Learning for Dynamics & Control Conference (L4DC 2026)</i>, Los Angeles, USA. (arXiv: 2602.16966) 4. S. Chakraborty*, A. K. Rege*, C. Monteleoni, L. Chen, BANDIT LEARNING ON DYNAMIC GRAPHS, <i>ARLET Workshop at Neural Information Processing Systems (NeurIPS 2025)</i>, San Diego, USA. 5. S. Chakraborty*, A. K. Rege*, C. Monteleoni, L. Chen, INCENTIVIZED LIPSCHITZ BANDITS, <i>IEEE Conference on Decision and Control (CDC 2025)</i>, Rio de Janeiro, Brazil. (arXiv: 2508.19466) 6. S. Chakraborty and L. Chen, INCENTIVIZED EXPLORATION IN NON-STATIONARY STOCHASTIC BANDITS, <i>IEEE American Control Conference (ACC 2024)</i>, Toronto, Canada. (arXiv: 2403.10819)
HONORS AND AWARDS	• Recipient, Best Paper Finalist (1 of 3; Top 1%), L4DC 2026. • Recipient, Summer Research Fellowship (2026). • Recipient, Graduate Research Assistant Fellowship (2026). • Recipient, Outstanding Teaching Assistant Award (CS Department). • Recipient, Outstanding Service Award (CS Department). • Recipient, Outstanding Research Paper Award (2024 & 2026 CS Dept.). • Recipient, Full Conference Travel Fellowship (IEEE ACC 2024 at Toronto, Canada and IEEE CDC 2025 at Rio de Janeiro, Brazil). • Recipient, CU Research Expo Poster Award (AY 2021–22 and AY 2023–24). • Recipient, Publication Recognition Award (AY 2023–24). • Recipient, Early Career Development Fellowship (CS Department). • Recipient, Lloyd Botway Award for Outstanding Master’s Student. • Selected as Lead Teaching Assistant (department lead; 2022–2025).
TEACHING	University of Colorado Boulder, Colorado, USA • Instructor of Record 2020 – 2021 • Introduction to Computational Thinking (~200 students) and Introduction to Data Science (~50 students). • Managed end-to-end course delivery, including lecture planning, assessments, and coordination of instructional staff. • CS Department Lead Teaching Assistant 2022 – 2025 • Selected via performance review to lead department-wide instructional quality. • Managed TA hiring, onboarding, and training of 280+ instructional staff. • Graduate Teaching Assistant 2020 – 2025 • Facilitated recitations, office hours, and grading for courses like Algorithms, Data Structures, and Introduction to Computing across 10+ semesters.
PROFESSIONAL SERVICE TECHNICAL SKILLS	Peer Reviewer: ICML, NeurIPS, IEEE CDC, ACC and TCNS (Journal). Languages: Python, C++, Java, Shell, R, Julia, JavaScript, MySQL Machine Learning: PyTorch, Pandas, NumPy, Matplotlib, Seaborn. Systems & Tools: Docker, Kubernetes, PySpark, Cascading, Flask, Django.
RELEVANT COURSEWORK	High-Dimensional Probability, Convex Optimization, Theory of Machine Learning, Deep Learning, Advanced Algorithms, Decision-Making Under Uncertainty, Algorithmic Game Theory.